STAFF REVIEW · H446 005 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q5

Do Now — source answers

1. 01101000 0011 6.5 = 110.1_2 = $0.1101_2 \times 2^3$. Mantissa 01101000 ; exponent 0011 .

2. 01101101_2 and 109 . 6 → 0110 , D → 1101 . In denary: (6 × 16) + 13 = 109 .

3. −10 −128 + 64 + 32 + 16 + 4 + 2 = −10 .

4. $00000110_2 = 06_{16}$ AND with 00001111 clears the upper nibble and preserves the lower nibble 0110 .

I Do — source worked steps

1. Evaluate the mantissa — CODE:
01011000
= ½ + ⅛ + ⅟₁₆
= 0.6875 — Read the mantissa as a signed binary fraction.

2. Convert the exponent to denary — CODE:
0011 = 3 — The exponent is a signed two’s-complement integer.

3. Apply the scale — CODE:
0.6875×2^3
= 0.6875 × 8
= 5.5

We Do — guided CHECK

1. Build the answer

2. Build the answer

3. Build the answer

4. Build the answer
Shift the mantissa left twice and reduce the exponent by 2.

You Do — source mark schemes

1. CODE:
Mantissa 0100 = 0.5
Exponent 1110 = -2

0.5 × 2^-2 = 0.125

Answer = 0.125

2. CODE:
Mantissa 1010 = -0.75

Exponent 1110 = -2

-0.75 × 2^-2 = -0.1875

Answer = -0.1875

3. CODE:
a) 010101 100 → normalised
b) 110101 111 → not normalised
c) 011010 010 → normalised
d) 101010 110 → normalised

Rule:
positive normalised mantissa begins 01
negative normalised mantissa begins 10

4. CODE:
(a) 01110000 0010

mantissa = 0.1110000_2 = 0.875
exponent = 2

0.875 × 2^2 = 3.5

(b) 10010000 1111

mantissa = -1 + 1/8 = -0.875
exponent = -1

-0.875 × 2^-1 = -0.4375

5. CODE:
More mantissa bits:
→ greater precision / more significant binary digits.

More exponent bits:
→ greater range / larger and smaller magnitudes can be represented.

If the total number of bits is fixed:
allocating more bits to the mantissa improves precision but leaves fewer bits for the exponent and reduces range;
allocating more bits to the exponent increases range but leaves fewer bits for the mantissa and reduces precision.

Exit Ticket — source answer

1. False. An exponent changes the magnitude; the sign of the mantissa determines whether the value is positive or negative.